



MINING DEALER BEST PRACTICE

# Cylinder Liner Projection Tool for 3500 Engine Block

Maintenance and  
Repair

Application

Component Life  
Management

Component  
Renewal (CRC)

MARC  
Management

|                                                      |    |
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**June 2023**  
**4.03-230607-2**

## 1.0 Introduction

This Best Practice was developed with the purpose of performing the test and optimal, effective, and reliable adjustment in Cylinder Liner Projection, guaranteeing its specification in accordance with the criteria CAT Cylinder Liner Projection and Head Bolt Torque Procedure for the Spacer Plate on Certain 3500 Engines (M0095046-02).

Due to the thickness change of the spacer plate, a new specification for allowable cylinder liner projection is published in Table 1.

| Part                                                             | Dimension                                    | Value |
|------------------------------------------------------------------|----------------------------------------------|-------|
| 519-7297 Spacer Plate                                            | 12.338 to 12.388 mm (0.48575 to 0.4877 inch) | X     |
| 454-9920 Cylinder Liner flange<br>211-7826 Cylinder Liner flange | 12.630 to 12.670 mm (0.4972 to 0.4988 inch)  | Z     |
| 144-5692 Plate Gasket                                            | 0.170 to 0.230 mm (0.0067 to 0.0091 inch)    | Y     |
| Allowable Liner Projection                                       | 0.050 to 0.120 mm (0.0020 to 0.0047 inch)    |       |

Table 1

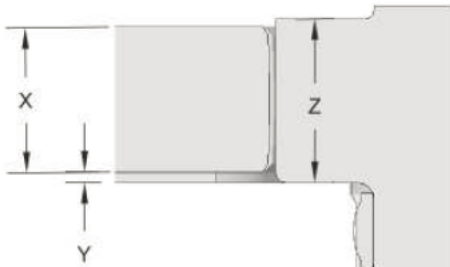


Illustration # 01 the measures that must be done in Allowable Liner Projection during the assembly process are observed.

## 2.0 Best Practice Description

In the Barloworld Mongolia machinery workshop, we developed a device that allows testing and adjustment in Allowable Liner Projection for CAT 3500 engine models, in a safe, fast, and reliable way, reducing the time of this test by 40%<.

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3.0 Implementation Steps

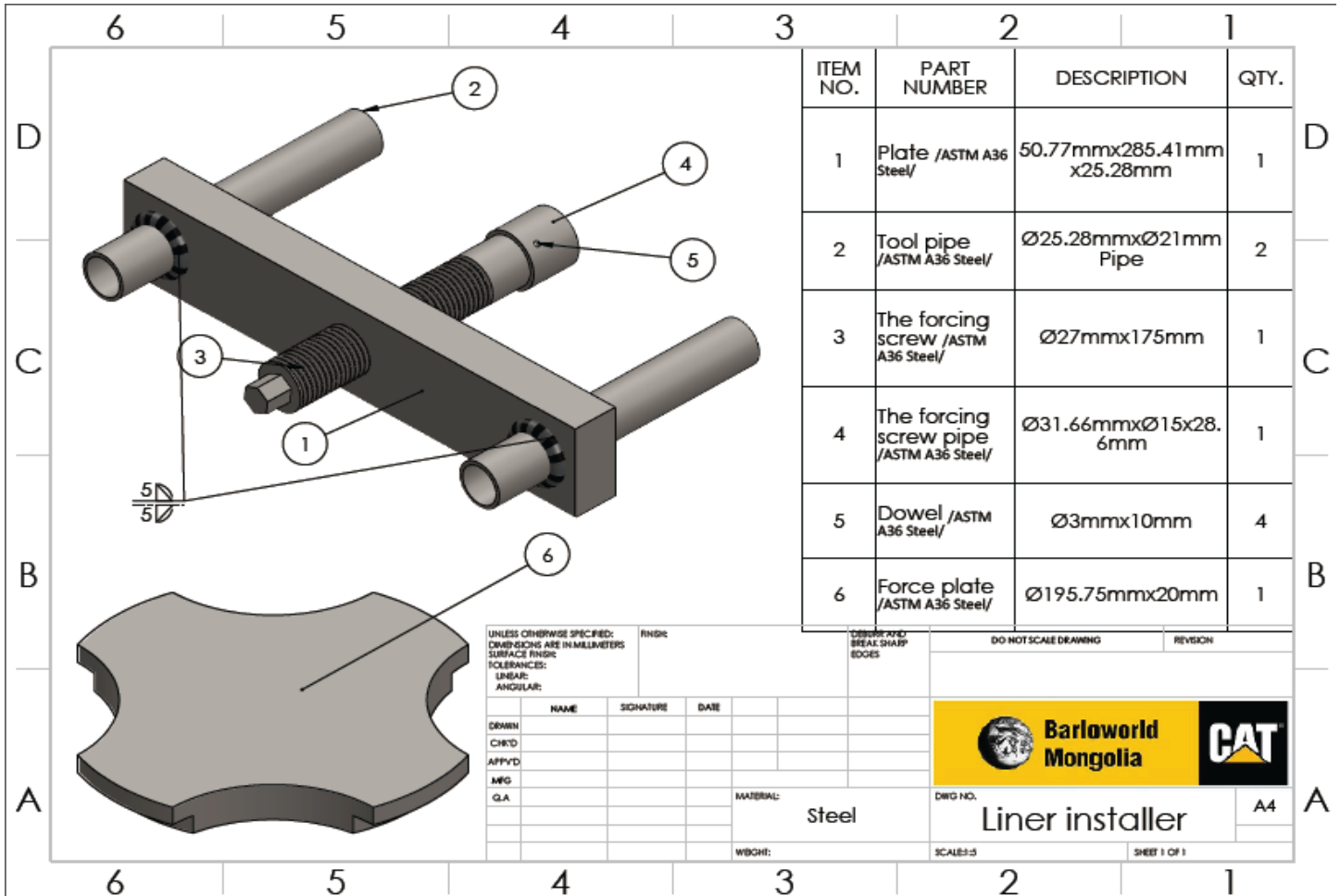
3.1 Manufacture of Cylinder Liner Projection Tool for block 3500, according to image and plans developed by the Barloworld Mongolia machining workshop, they will be able to manufacture this device that allows testing more efficiently.



Illustration # 01- and 02-Cylinder Liner Projection Tool is observed.

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# Material's list



01 of 06

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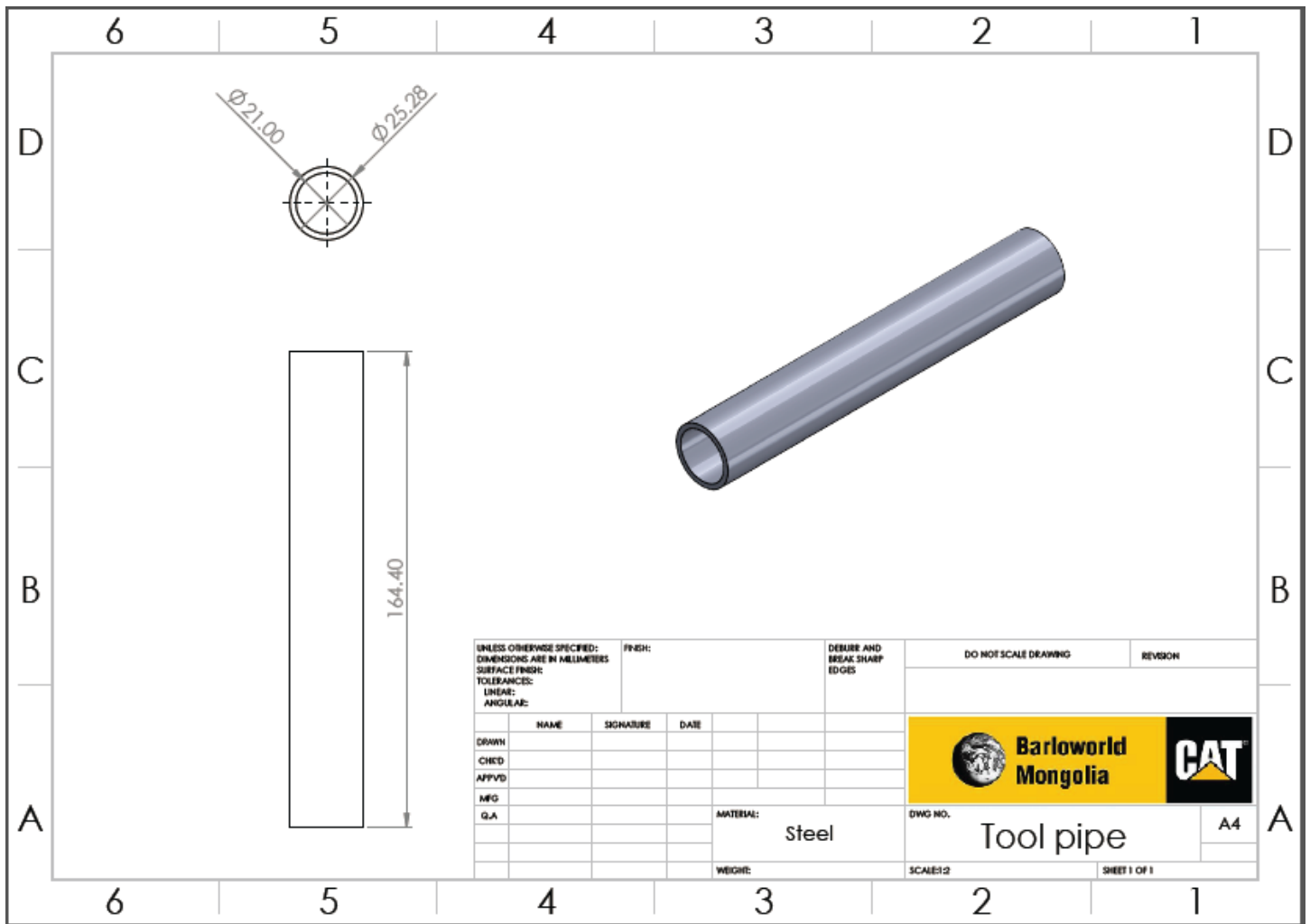
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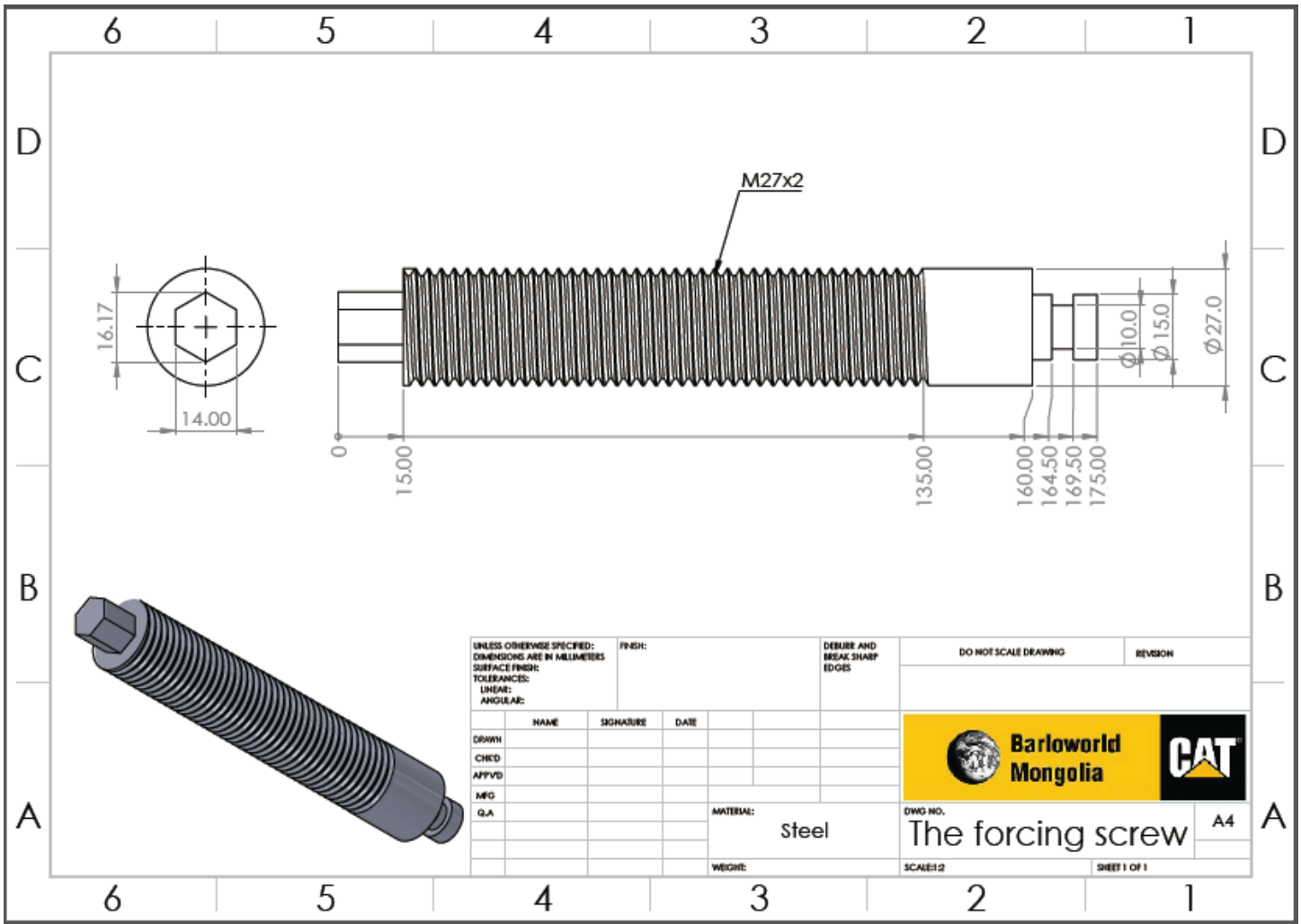
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Cylinder Liner Projection Tool for 3500 Engine Block

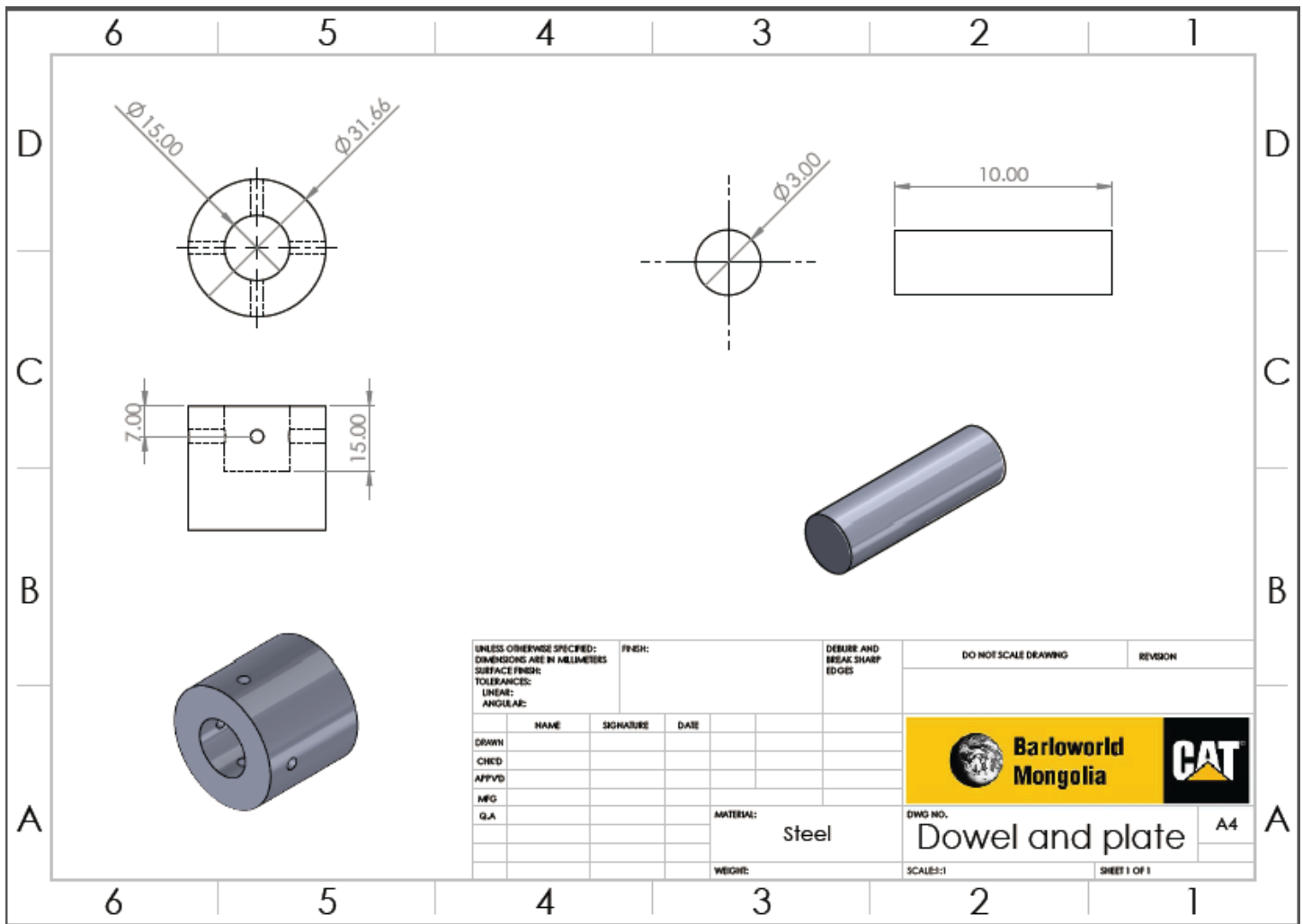
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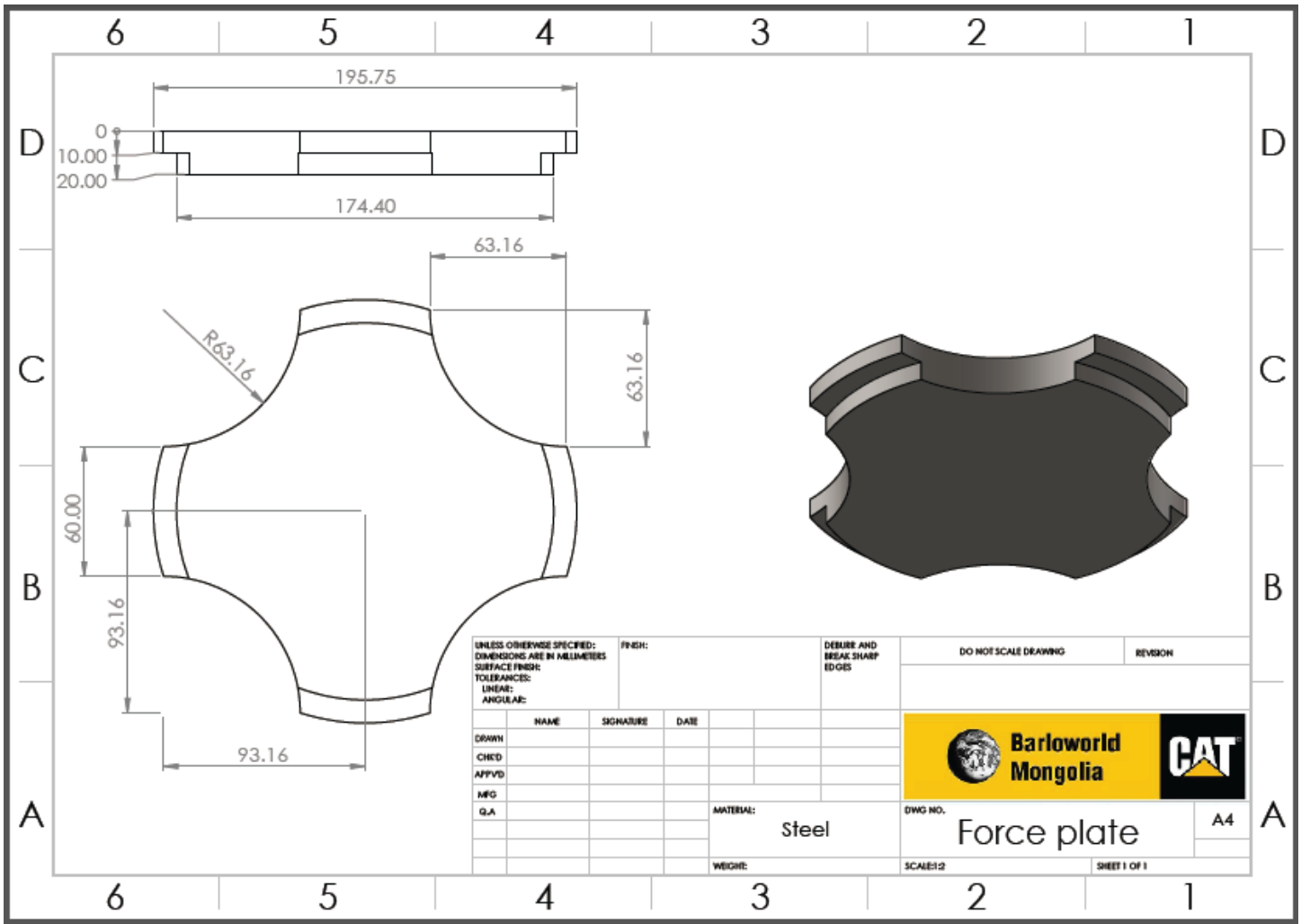
Cylinder Liner Projection Tool for 3500 Engine Block

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3.2 perform test and adjust according to Cylinder Liner Projection (SENR1123-21) prepare Cylinder Liner Projection Tool and the following tools.

|         |                                                  |   |
|---------|--------------------------------------------------|---|
| 8F-6123 | Bolt (3/4-16 thread, 139.7 mm (5.5.50 inch) long | 2 |
| 3B-1925 | Washer (COPPER)                                  | 6 |
| 1A-0075 | Bolt (3/4-16 thread, 44.45 mm (1.750 inch) long  | 6 |
| 8T-0455 | Liner Projection Tool Group                      | 1 |
| 6V-7059 | Micrometer                                       | 1 |

Liner Projection Tool



Illustration #06

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3.3 Torque process for test place the copper washers 3B-1925 and bolt 8F-6123 and 1A-0075.

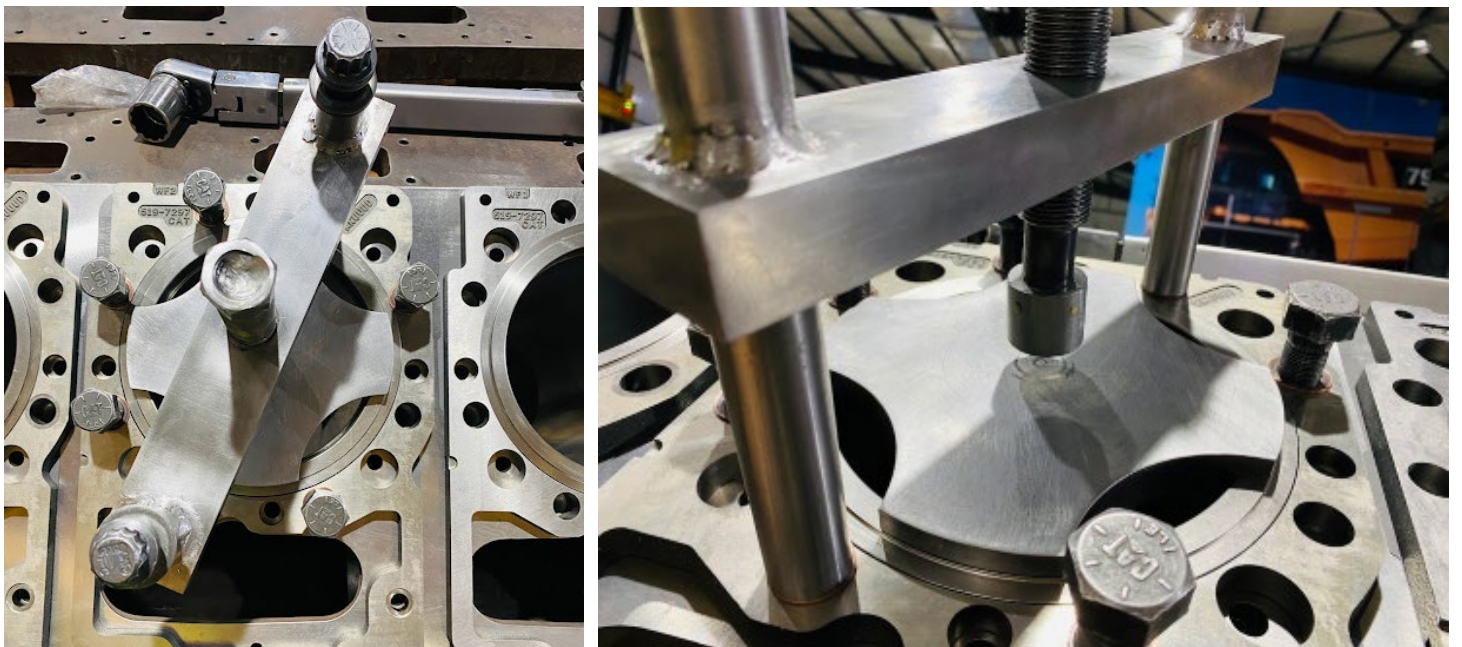
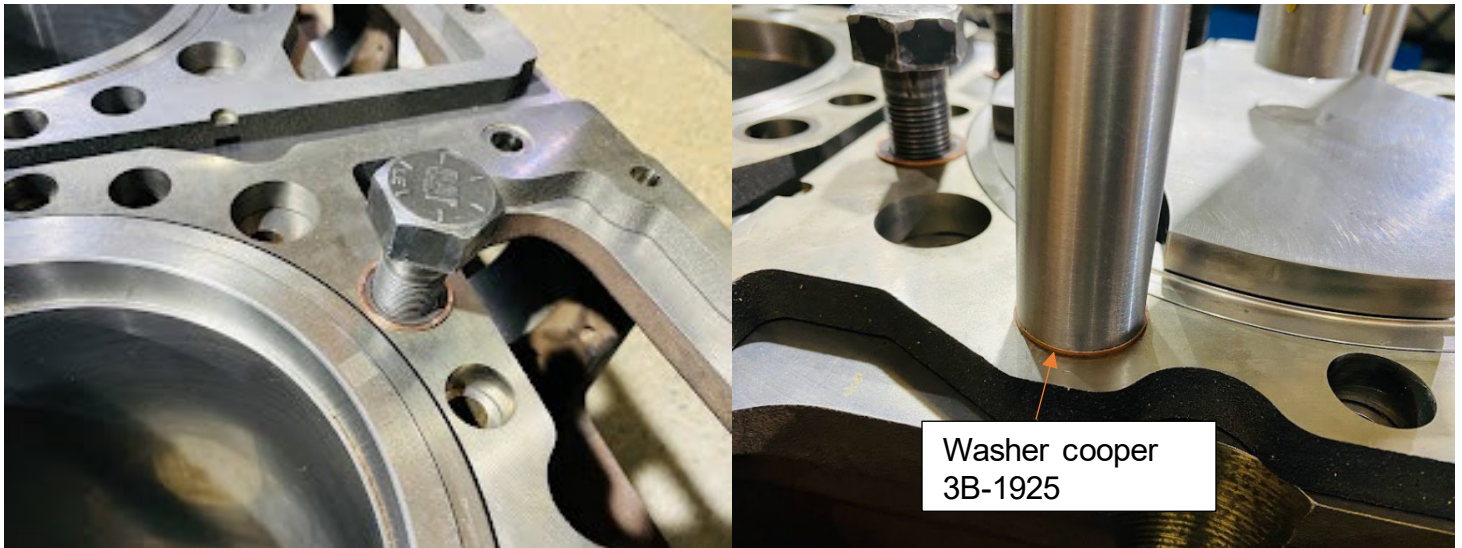


Illustration # 07,08,09 and 10 place the copper washers 3B-1925 and bolt 8F-6123 and 1A-0075.

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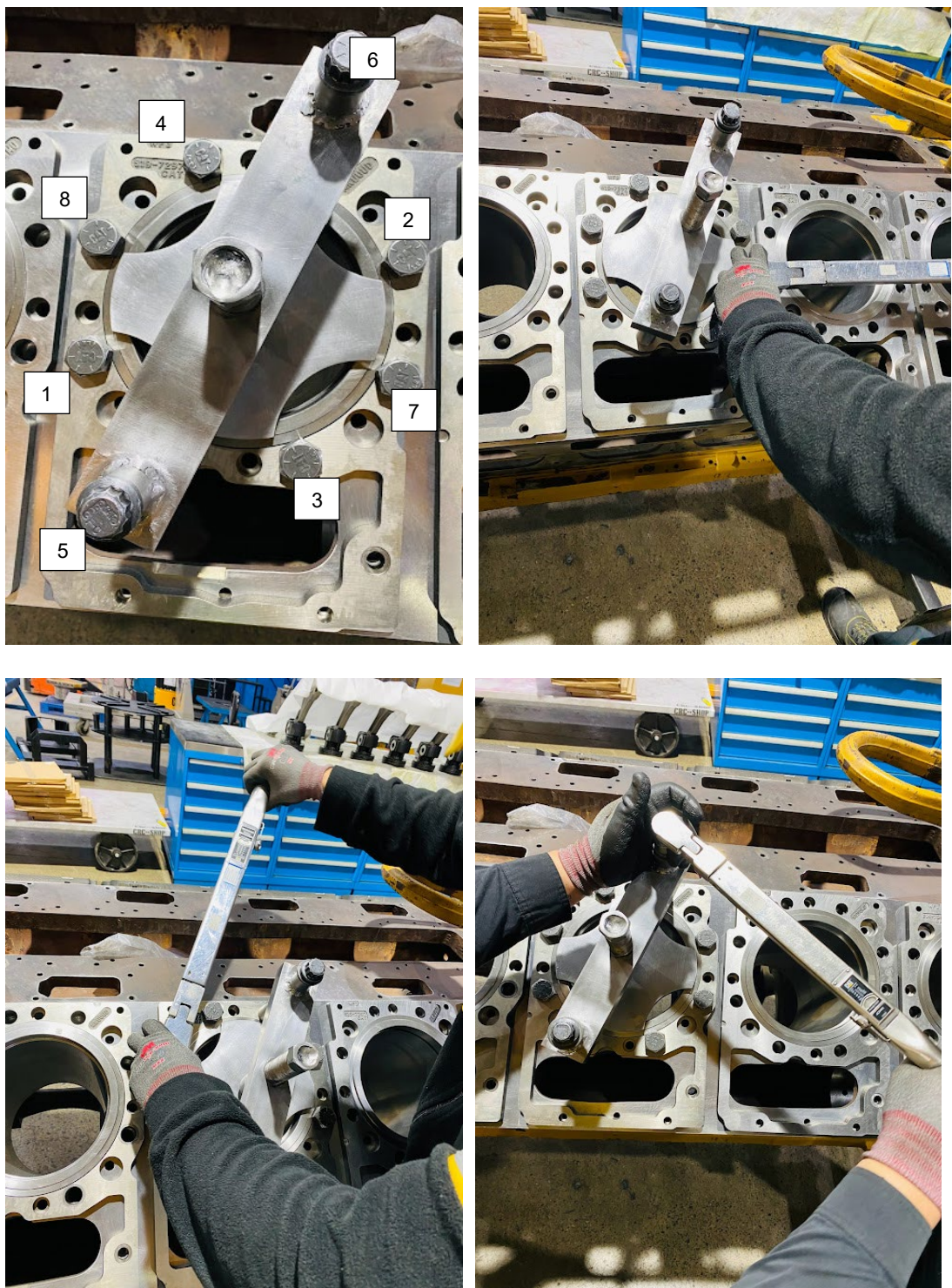


Illustration #11,12,13 and 14 bolt tightening torque is 95 Nm (70lb ft) in the numbered order in illustration #11.

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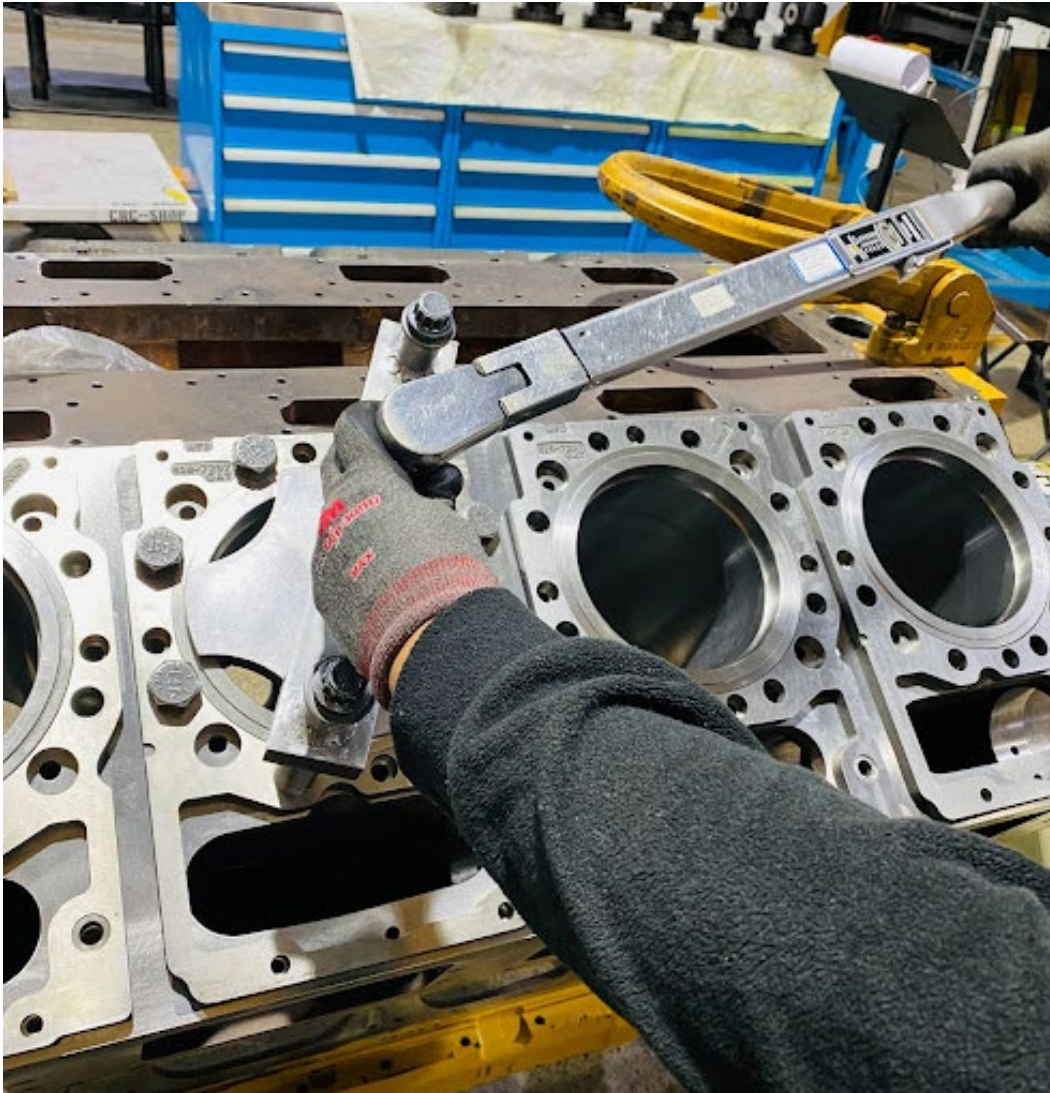
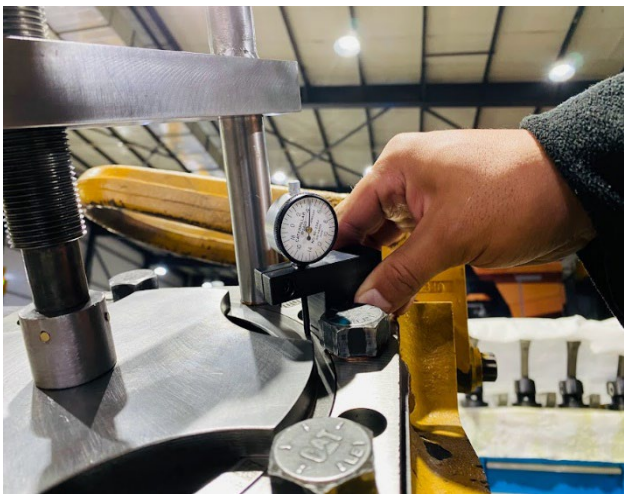
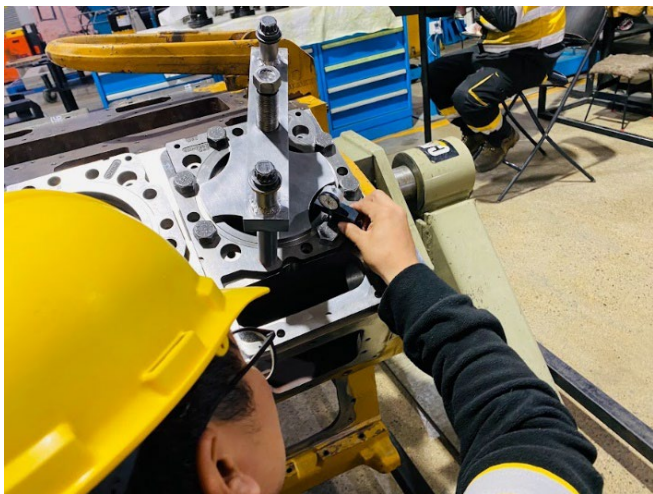


Illustration #15 the torque is applied to the central bolt of the tool torque of 70 Nm (50lb ft) that pushes the plate.

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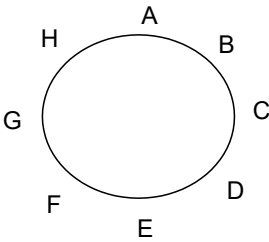
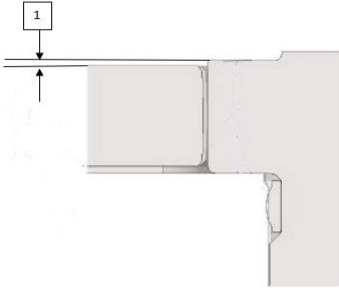
Make projection measurements in an easy, safe, and reliable way and record on quality sheet.

|                                                                                                                 |                  |                 |                         |
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3.2 measurement record attached quality sheet used by Barloworld Mongolia which must mark what type of plate is used for the specification.

|                                                                                   |  |                                                  |  |
|-----------------------------------------------------------------------------------|--|--------------------------------------------------|--|
|  |  | EVALUATION IN ALLOWABLE LINER PROJECTION CAT3500 |  |
| DATE                                                                              |  | SUB-COMPONENTE:                                  |  |
| MS WO.: CRC WO                                                                    |  | BRAND:                                           |  |
| CUSTOMER:                                                                         |  | MODEL:                                           |  |
| TECNICAL:                                                                         |  | SERIAL:                                          |  |

PROJECTION

ALLOWABLE LINER PROJECTION

|    | A | B | C | D | E | F | G | H |
|----|---|---|---|---|---|---|---|---|
| 1  |   |   |   |   |   |   |   |   |
| 2  |   |   |   |   |   |   |   |   |
| 3  |   |   |   |   |   |   |   |   |
| 4  |   |   |   |   |   |   |   |   |
| 5  |   |   |   |   |   |   |   |   |
| 6  |   |   |   |   |   |   |   |   |
| 7  |   |   |   |   |   |   |   |   |
| 8  |   |   |   |   |   |   |   |   |
| 9  |   |   |   |   |   |   |   |   |
| 10 |   |   |   |   |   |   |   |   |
| 11 |   |   |   |   |   |   |   |   |
| 12 |   |   |   |   |   |   |   |   |
| 13 |   |   |   |   |   |   |   |   |
| 14 |   |   |   |   |   |   |   |   |
| 15 |   |   |   |   |   |   |   |   |
| 16 |   |   |   |   |   |   |   |   |

INSPECTION DIMENSIONAL

| Allowable Liner Projection Spacer Plate 519-7297 |             |         |              |         | Allowable Liner Projection Spacer Plate 362-9677 |             |         |              |         |
|--------------------------------------------------|-------------|---------|--------------|---------|--------------------------------------------------|-------------|---------|--------------|---------|
| Parte                                            | Minimun(mm) | Max(mm) | Minimun (in) | max(in) | Parte                                            | Minimun(mm) | Max(mm) | Minimun (in) | max(in) |
| 1                                                | 0.050       | 0.120   | 0.002"       | 0.0047" | 1                                                | 0.100       | 0.170   | 0.004"       | 0.007"  |
|                                                  |             |         |              |         |                                                  |             |         |              |         |
|                                                  |             |         |              |         |                                                  |             |         |              |         |

**SUMARY:**

**APPROVED FOR :**



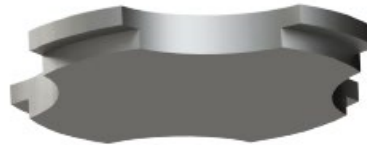
## 4.0 Benefits

- 4.1 Improved component handling
- 4.2 Greater efficiency in the process
- 4.3 Reduce test time by 40% less time
- 4.4 Measurement reliability

## 5.0 Resources Required

|         |                                                  |   |
|---------|--------------------------------------------------|---|
| 8F-6123 | Bolt (3/4-16 thread, 139.7 mm (5.5.50 inch) long | 2 |
| 3B-1925 | Washer (COPPER)                                  | 4 |
| 1A-0075 | Bolt (3/4-16 thread, 44.45 mm (1.750 inch) long  | 4 |
| 8T-0455 | Liner Projection Tool Group                      | 1 |
| 6V-7059 | Micrometer                                       | 1 |

### Fabrication Cylinder Liner Projection Tool



### 5.1 Estimated Costs

| Machine           | Time            |
|-------------------|-----------------|
| Horizontal Lathe  | 8hr             |
| Center Machining  | 5hr             |
| Polish & Cleaning | 1hr             |
| Material          | 70\$USD         |
| <b>Total Cost</b> | <b>380\$USD</b> |

## 6.0 Supporting Attachments / References

6.1 Cylinder Liner Projection and Head Bolt Torque Procedure for the 519-7297 Spacer Plate on Certain 3500 Engines {1100, 1201,1221} (M0095046-02)

6.2 Cylinder Liner Projection (SENR1123-21)

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## 7.0 Related Best Practices

None

## 8.0 Acknowledgements

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 Pablo Salazar / Machine Shop Manager

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